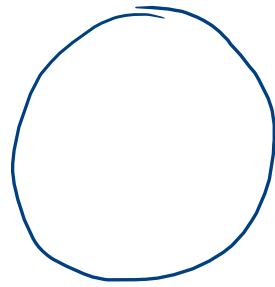
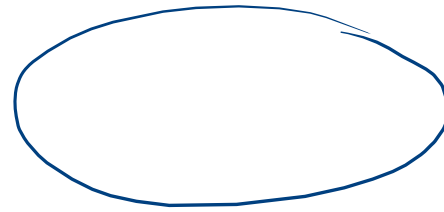


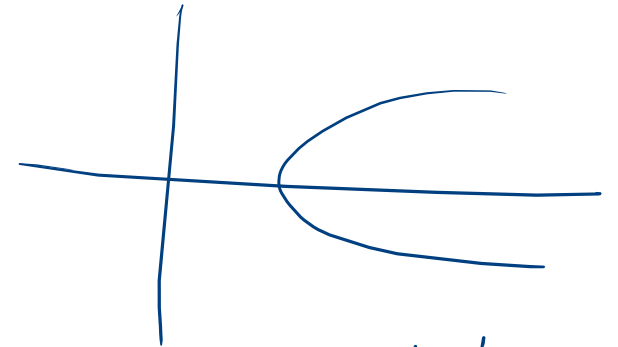
Conic Section:



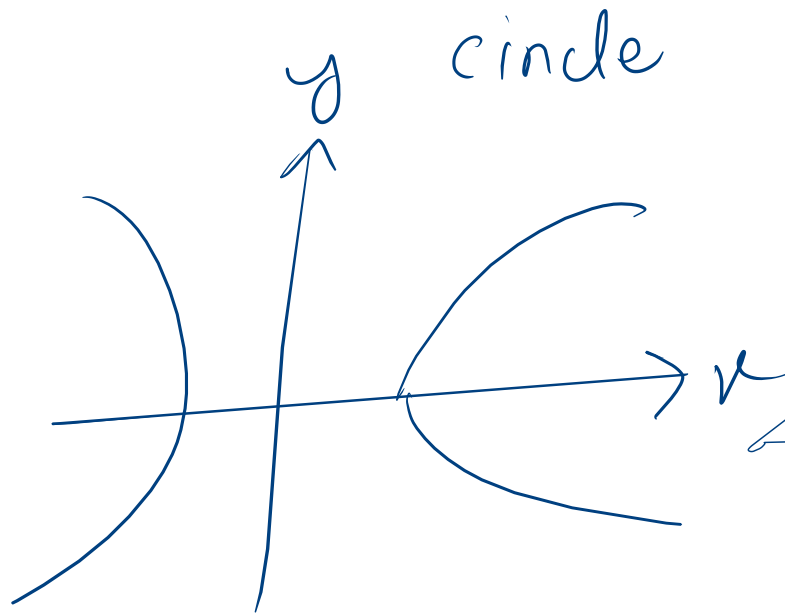
circle



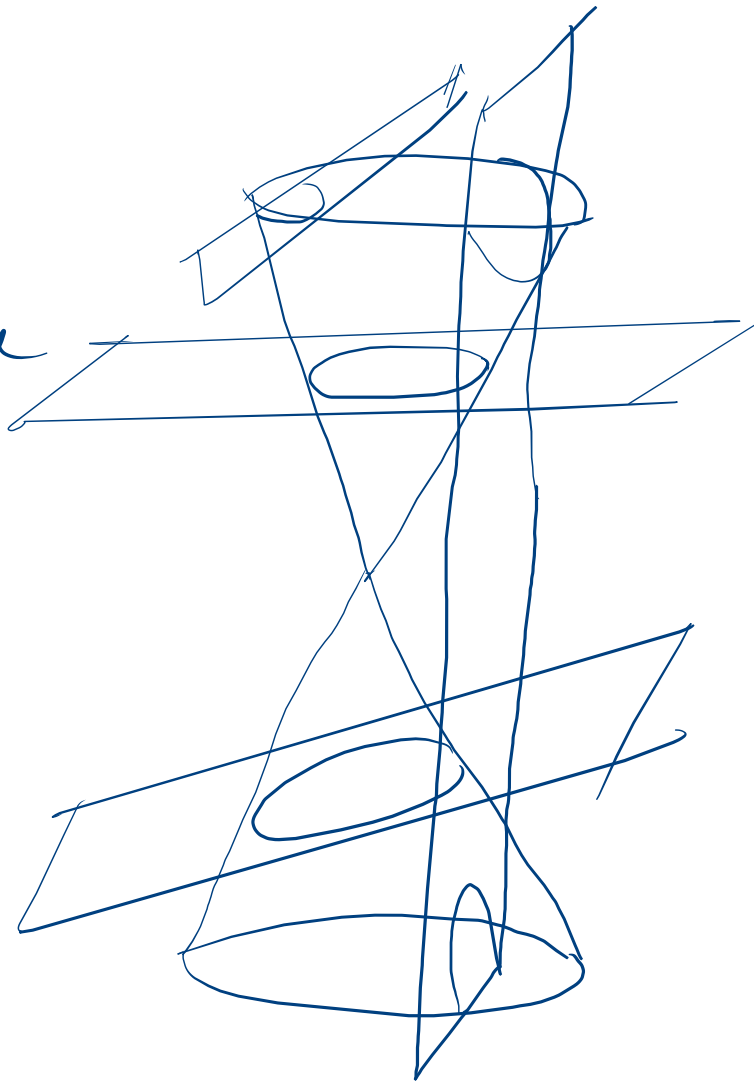
Ellipse



parabola



Hyperbola



$$\boxed{z^2 = x^2 + y^2}$$

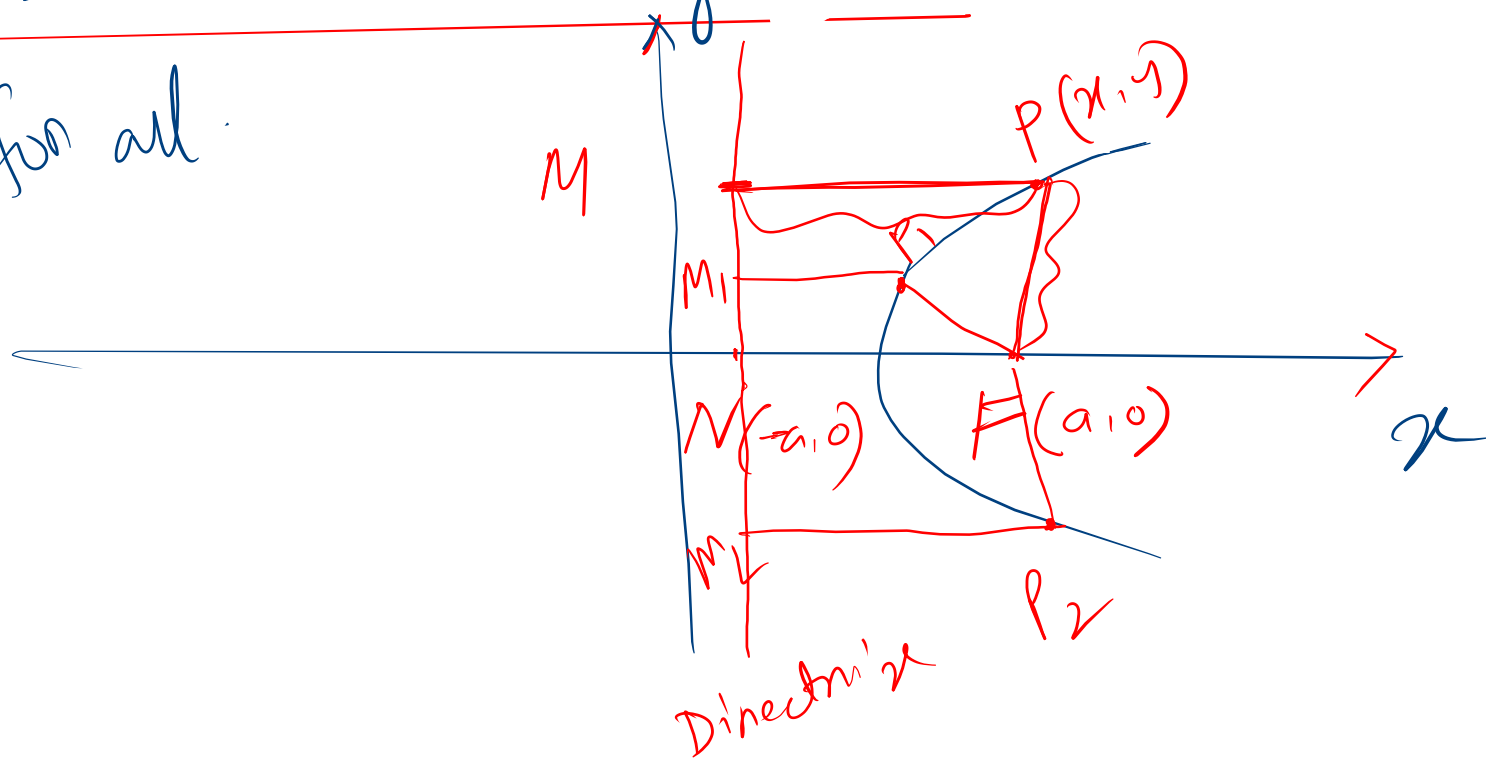
cone

$$z = \pm \sqrt{x^2 + y^2}$$
$$f(x, y) = \pm \sqrt{x^2 + y^2}$$

parabola: A parabola is the set of points in a plane that are equidistance from a fixed point F which is called focus and a fixed line which is called Directrix.

$$S_1 = \{ \underline{(x, y)} \mid \underline{FP = PM}, \forall (x, y) \in S_1 \}$$

$\forall$ for all.

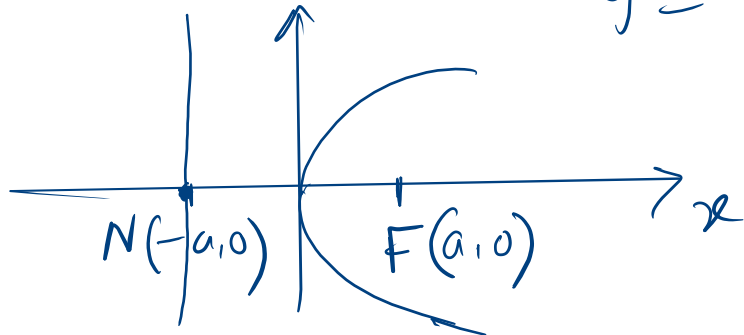


Parabola

$$y^2 = 4ax$$

$$a > 0$$

$$y^2 = 4x, a = 1,$$



Center $C(0,0)$

Focus $F(a,0)$ Eqn of Focus, $x = a$

Directrix point $N(-a,0)$, Eqn of Directrix,

Eccentricity $e = 1$

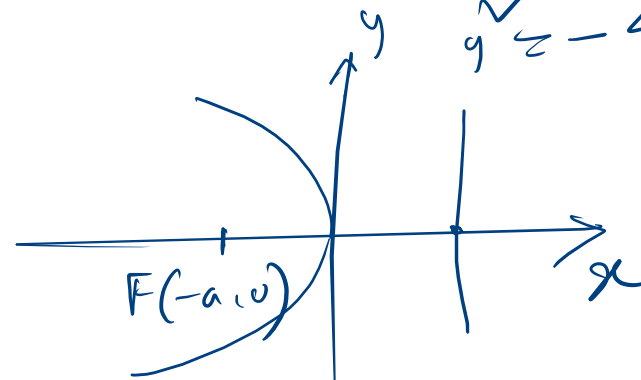
Latus rectum $= 4/a$ unit.

$$x = -a$$

(Note: Parabola always opens its mouth towards the axis which is one degree).

$$y^2 = 4ax, a < 0$$

$$y^2 = -4x, a = -1.$$



Center = Vertex

center $(0,0)$

$F(-a,0)$, Eqn of focus $x = -a$

Directrix, $N(a,0)$, Eqn of Directrix $x = a$.

$e = 1$

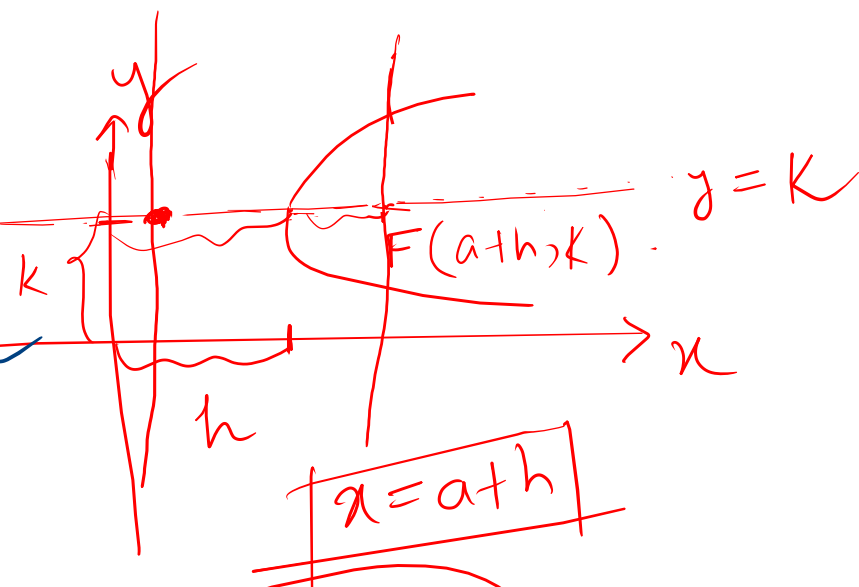
Latus rectum $= 4/a$ unit.

~~(*)~~ $(y-k)^2 = 4a(x-h)$, $h, k > 0$.

1. Center $C(h, k)$.

Standard form of parabola

2. Focus $F(a+h, k)$



$$\underline{x-h=a}$$

$$y-k=0$$

$$\Rightarrow x=a+h$$

$$\Rightarrow y=k$$

Eqn of focus, & latus rectum $x=a+h$

3. Eccentricity, $e=1$.

4. latus rectum = $4|a|$ unit.

5. Directrix point on the axis $(h-a, k)$

$$x-h=-a, y=k$$

$$x=h-a$$

Eqn of Directrix, $x=h-a$.

* $(x-h)^2 = 4a(y-k)$, $a, h, k > 0$,
 Standard form of parabola.

1. center $c(h, k)$.
2. focus $F(h, k+a)$.

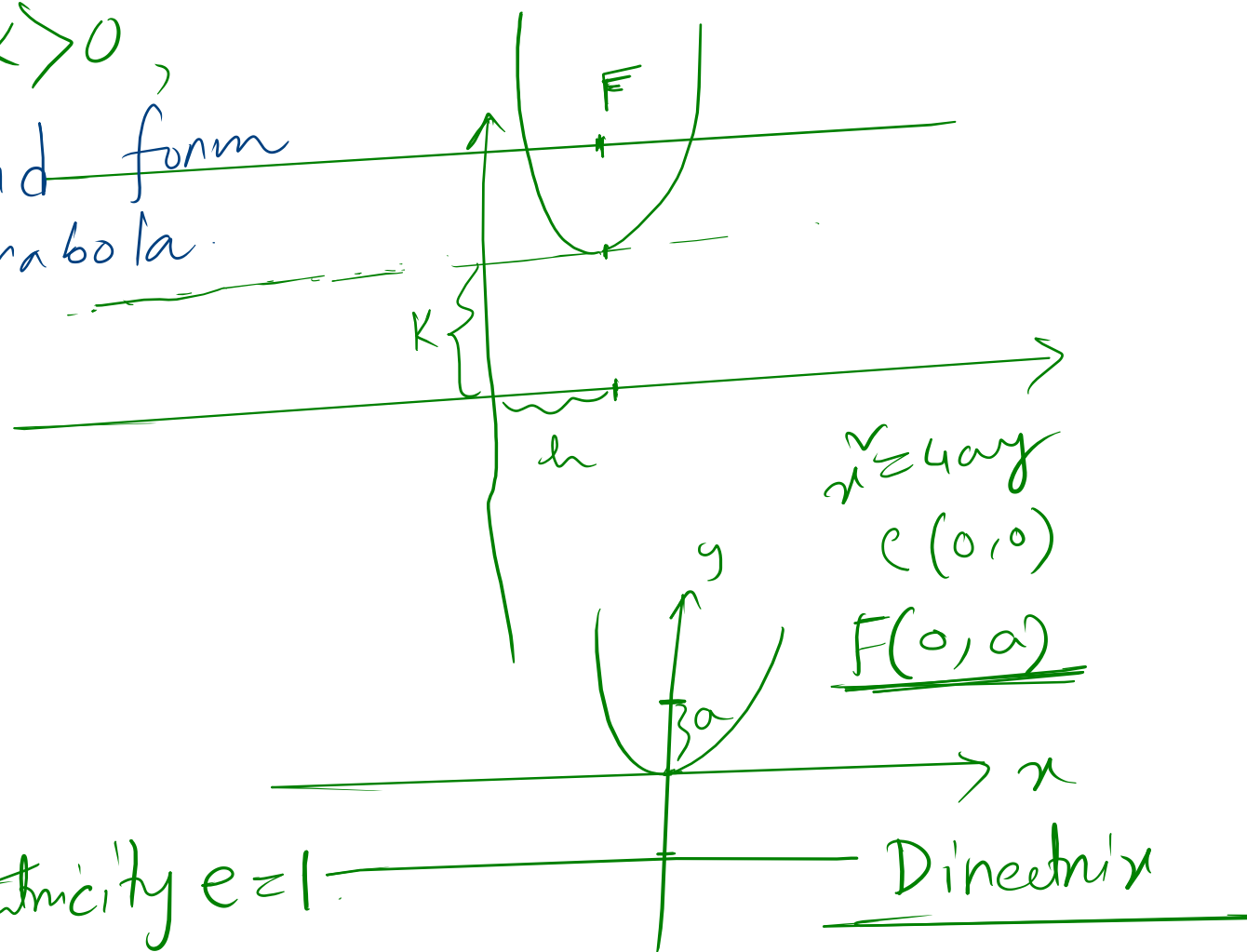
$$x-h=0 \quad y-k=a$$

$$\Rightarrow x=h \quad \Rightarrow y=k+a$$

Equation of $y=k+a$
 focus & latus rectum

3. latus rectum, $4|a|$ unit, Eccentricity $e=1$.

4. Directrix point, $(h, k-a)$, Equation of Directrix, $y=k-a$



Ex: 01 Transform the conics $y^2 + 6y + 2x + 1 = 0$ into standard form, then find center, vertex, focus, directrix, length of latus rectum, eccentricity of conics, equation of latus rectum, identify conics & sketch it.

$$y^2 + 6y + 2x + 1 = 0$$

$$\Rightarrow y^2 + 6y = -(2x + 1)$$

$$\Rightarrow y^2 + 6y + 9 = -2x - 1 + 9$$

$$\Rightarrow (y + 3)^2 = -2(x - 4)$$

$$\Rightarrow (y + 3)^2 = 4 \left(-\frac{1}{2} \right) (x - 4)$$

1. C (4, -3), V (4, -3), Focus, $F\left(\frac{7}{2}, -3\right)$

$$4a = -2$$

$$\Rightarrow a = -\frac{1}{2}$$

length of Latus rectum.
 $4|a| = 2$ unit.

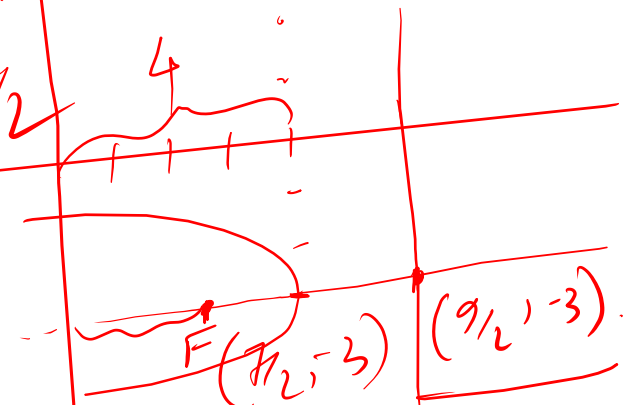
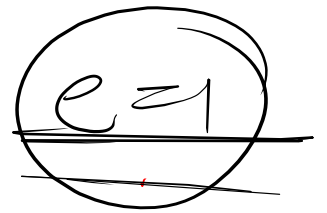
Eqn of Latus rectum, $x = \frac{7}{2}$

$$x - 4 = -\frac{1}{2}$$

$$x = -\frac{1}{2} + 4$$

$$y = -3$$

Eqn of Directrix, $x - 4 = \frac{1}{2}$
 $\Rightarrow x = 4 + \frac{1}{2} = \frac{9}{2}$



$$* 2x^2 - 16x - 3y + 38 = 0.$$

$$\Rightarrow 2x^2 - 16x = 3y - 38$$

$$\Rightarrow 2(x^2 - 8x) = 3y - 38$$

$$\Rightarrow 2(x^2 - 2 \cdot 4x + 16) = 3y - 38 + 32$$

$$\Rightarrow 2(x-4)^2 = 3y - 6$$

$$\Rightarrow 2(x-4)^2 = 3(y-2)$$

$$\Rightarrow (x-4)^2 = \frac{3}{2}(y-2)$$

$$\Rightarrow (x-4)^2 = 4 \cdot \frac{3}{8}(y-2)$$

$$C(4,2), V(4,2), \text{Focus} \left(4, \frac{19}{8}\right),$$

$$e = 1.$$

$$y-2 = \frac{3}{8}$$

$$\Rightarrow y = \frac{3}{8} + 2$$

$$\begin{cases} 4a = 3/2 \\ \Rightarrow a = \frac{3}{8} \end{cases}$$

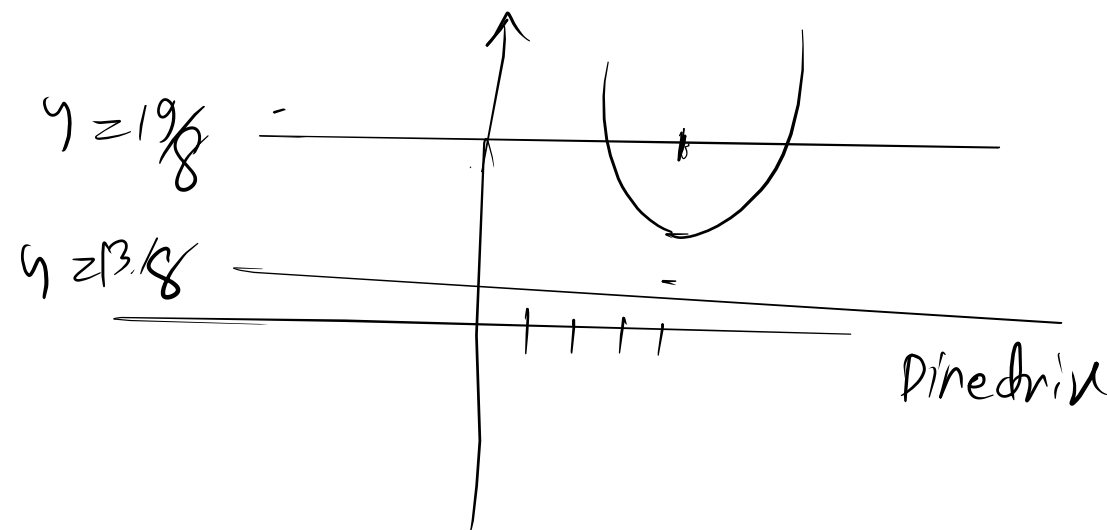
$$\text{later rectum } 4 \cdot \frac{3}{8} \text{ unit} = \frac{3}{2} \text{ unit}$$

$$\text{Equation of later rectum, } y = \frac{19}{8}$$

$$\text{Directrix point } \left(4, \frac{13}{8}\right)$$

$$y-2 = -\frac{3}{8}$$

$$\Rightarrow y = -\frac{3}{8} + 2$$



Far of

Directrix

$$y = \frac{13}{8}$$